Exploring Token Serialization for Mamba-Based LiDAR Point Cloud Segmentation

Dening Lu, Kyle Gao, *Member, IEEE*, Jonathan Li, *Fellow, IEEE*, Dedong Zhang, *Member, IEEE*, and Linlin Xu, *Member, IEEE*

***Abstract*—LiDAR point cloud segmentation has increasingly benefited from the application of Mamba-based models. However, the unordered and irregular natures of point clouds necessitate serialization, which significantly impacts the performance of Mamba-based methods. This article explores the critical role of token serialization in Mamba-based point cloud processing, using the pure Mamba network, PointMamba, as the baseline. We systematically investigated existing point cloud serialization methods, evaluating their performance on two challenging LiDAR datasets: the airborne MultiSpectral LiDAR (MS-LiDAR) dataset and the aerial Dayton annotated LiDAR Earth scan (DALES) dataset. To explore the inherent factors of serialization contributing to Mamba's performance, we design novel indicators for serialization quality, focusing on spatial and semantic proximity. These indicators are validated across all datasets, offering a valuable reference and guidance for advancing token serialization in Mamba-based point cloud processing. Guided by these indicators, we proposed a new point cloud serialization method that integrates spatial and semantic features through a weighted comprehensive distance matrix. The proposed method achieves superior accuracy on both LiDAR datasets, surpassing existing approaches, and establishes a strong foundation for advancing Mamba-based point cloud processing.**

***Index Terms*—Deep learning, LiDAR point cloud segmentation, Mamba, token serialization.**

I. Introduction

As a state-of-the-art (SOTA) technique of state-space models (SSMs), Mamba has recently emerged as a highly promising method with great performance in long-range dependency modeling [1]. Given the great success it achieved in natural language processing (NLP) and image processing [2], [3], there are more and more works [4], [5], [6] proposed to explore the applications of Mamba in point cloud processing.

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Dening Lu and Kyle Gao are with the Department of Systems Design Engineering, University of Waterloo, Waterloo, ON N2L 3G1, Canada (e-mail: d62lu@uwaterloo.ca; y56gao@uwaterloo.ca).

Jonathan Li is with the Department of Systems Design Engineering and the Department of Geography and Environmental Management, University of Waterloo, Waterloo, ON N2L 3G1, Canada (e-mail: junli@uwaterloo.ca).

Dedong Zhang is with the Department of Geography and Environmental Management, University of Waterloo, Waterloo, ON N2L 3G1, Canada, and also with the Tianfu Jiangxi Laboratory, Chengdu, Sichuan 641419, China (e-mail: dedong.zhang@uwaterloo.ca).

Linlin Xu is with the Department of Geomatics Engineering, University of Calgary, Calgary, AB T2N 1N4, Canada (e-mail: lincoln.xu@ucalgary.ca).

Data is available on-line at https://github.com/d62lu/Mamba-Serialization

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Adapting Mamba to point cloud processing presents numerous challenges due to the significant differences between point clouds and natural language or pixel-based data, including sparsity, uneven spatial distribution, and disorder. These differences necessitate an in-depth discussion of Mamba's adaptability to point cloud processing.

First, the sparsity and uneven spatial distribution of point clouds make local and global feature extraction particularly challenging. Mamba's selective state-space model addresses this issue by dynamically adjusting its state-space parameters based on the input sequence. This adaptability allows the model to analyze each input point, determining whether to propagate or discard past information. Second, Mamba's selective state-space model integrates contextual information while filtering out irrelevant or redundant data. This capability is especially advantageous for handling noise and outliers in point clouds, such as those present in LiDAR scans, making it well-suited for noisy data processing. Recent Mamba-based methods, such as PointMamba [6] and PointCloudMamba [4], have demonstrated the model's effectiveness in processing both synthetic and real point cloud datasets. Finally, the unordered nature of point clouds makes serialization a critical step for processing with Mamba. Existing point cloud serialization methods typically utilize the geometric spatial attributes of point clouds to establish an ordering that preserves the original spatial topology and geometric relationships during serialization.

Given the critical role of point cloud serialization in Mamba applications, it is essential to analyze the impact of various serialization methods on Mamba's performance and explore the underlying factors contributing to performance. To the best of our knowledge, there is no relevant overview and analysis of point cloud serialization for Mamba-based methods.

To fully explore the effect of token serialization to Mamba's performance in LiDAR point cloud processing, this article takes a pure Mamba network, PointMamba [6], as a baseline, investigating and analyzing various existing point cloud serialization methods. Furthermore, we explored the inherent factors of point cloud serialization contributing to Mamba's performance, which leads to an optimized point cloud serialization method for better performance of Mamba in LiDAR point cloud processing.

The main contributions of our work are summarized as follows.

1) Taking a pure Mamba network, PointMamba [6], as a baseline, we investigated and analyzed seven existing

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token serialization methods of Mamba in LiDAR point cloud processing. Their performance was tested on two challenging LiDAR datasets (airborne MultiSpectral LiDAR (MS-LiDAR) dataset and aerial Dayton annotated LiDAR Earth scan (DALES) dataset), to eliminate experimental contingencies and ensure the robustness of the comparative analysis.

2) We proposed robust and reliable indicators to evaluate the serialization quality, exploring the inherent factors of point cloud serialization contributing to Mamba's performance. The effectiveness of the proposed indicators has been confirmed on both LiDAR datasets. It provides a valuable reference and guidance for advancing token serialization in Mamba-based point cloud processing.
3) Guided by the proposed indicators for serialization quality evaluation, we proposed a novel point cloud serialization method, geometric–semantic fusion (GSF) serialization, effectively incorporating both spatial and semantic proximity. GSF outperforms existing serialization methods, achieving superior results on two benchmark LiDAR datasets.

II. Related Work

A. LiDAR Point Cloud Segmentation

LiDAR point cloud segmentation techniques can generally be categorized into four primary approaches: volume-based methods, projection-based methods, point-based methods, and attention-based methods.

Volume-based methods, inspired by image processing, involve converting point clouds into structured representations. VoxNet [7] pioneered this approach by using a 3-D voxel grid to organize unstructured point cloud data, enabling the application of 3-D convolutions for feature extraction. To improve the efficiency, OctNet [8] introduced an unbalanced grid-octree structure, allowing higher resolution inputs (e.g., $256 \times 256 \times 256$) compared to VoxNet. Later advancements, such as 4-D Spatio-temporal ConvNets [9], employed sparse convolution techniques to eliminate redundant computations in empty voxels, significantly reducing memory and computational demands typically associated with voxel-based methods.

Projection-based methods simplify 3-D point cloud processing by transforming the data into 2-D representations, facilitating the use of established CNN architectures. For instance, SqueezeSeg [10] and its later iterations (SqueezeSegV2 [11] and SqueezeSegV3 [12]) leverage spherical or range projections for grid-based semantic segmentation. RangeNet++ [13] converts point clouds into range images, processes them with 2-D fully convolutional networks, and reconstructs the 3-D structure after segmentation. Similarly, PolarNet [14] and multiprojection fusion (MPF) [15] explored alternative projection strategies, such as polar and multiview projections, to enhance segmentation accuracy. Despite their computational efficiency, they face challenges like discretization errors and difficulties in handling complex geometric structures.

Point-based methods directly operate on raw point cloud data, preserving the unordered nature of the input. PointNet [16] introduced shared MLPs for learning point-wise features, and PointNet++ [17] enhanced this by incorporating local feature aggregation. Building on these ideas, methods like PointCNN [18], PointConv [19], and DGCNN [20] utilized convolutional and graph-based operations to improve local feature extraction. Recent work such as PointNext [21] has further refined point-based methods, exploring data augmentation strategies to boost segmentation performance. PointMLP [22], by contrast, adopted a lightweight MLP-based architecture with residual connections to efficiently extract hierarchical features and achieve strong results in point cloud segmentation.

The representative work of the attention-based approach is the Transformer. Inspired by their success in computer vision, Transformers have gained prominence in point cloud segmentation. The self-attention mechanism of Transformers excels at capturing global context. Point cloud Transformer (PCT) [23] replaced PointNet's MLPs with Transformer blocks to model long-range dependencies. Point Transformer (PT) [24] focused on applying Transformers for local feature extraction, demonstrating superior aggregation capabilities. Instead of raw point cloud input, Stratified Transformer [25] utilized voxelized data and implemented Transformers within local windows, following the Swin Transformer paradigm [26]. Efficient Transformers such as PPT-Net [27], PatchFormer [28], and SPFormer [29] have been developed to address computational challenges. PPT-Net employed a hierarchical encoder–decoder structure, combining graph convolution-based local embedding with Transformer-based global feature learning to reduce computational costs. Superpoint-based methods like PatchFormer and SPFormer created geometrically homogeneous clusters as a preprocessing step, significantly reducing the number of computations while maintaining segmentation accuracy.

In addition, Mamba also belongs to attention-based methods, which is a type of linear attention model [30]. The detailed introduction of Mamba is shown in Section II-B.

B. Mamba-Based Point Cloud Processing

Mamba [1] was proposed to further solve the limitation of the inability to perform content-based reasoning in previous SSMs [31], [32], [33]. Its system parameters are determined by the input, which enables the model to adaptively select information along the sequence length dimension depending on the current token.

Recently, the application of Mamba in point cloud analysis has emerged. Point cloud Mamba (PCM) [4] designed a hierarchical structure combining a local feature extraction module and Mamba-based global context modeling for point cloud processing, achieving excellent performance on datasets like S3DIS [34] and ScanObjectNN [35]. Serialized Point Mamba [36] leveraged grid pooling and conditional positional encoding to integrate local–global modeling, showing notable efficiency improvements. PointMamba [6] focused on global modeling by utilizing space-filling curves like the Hilbert and Trans-Hilbert curves for point tokenization, reducing computational costs while delivering competitive performance across multiple benchmarks. PoinTramba [5] combined Mamba's efficiency with Transformer's global modeling power for enhanced point cloud analysis, introducing a bi-directional

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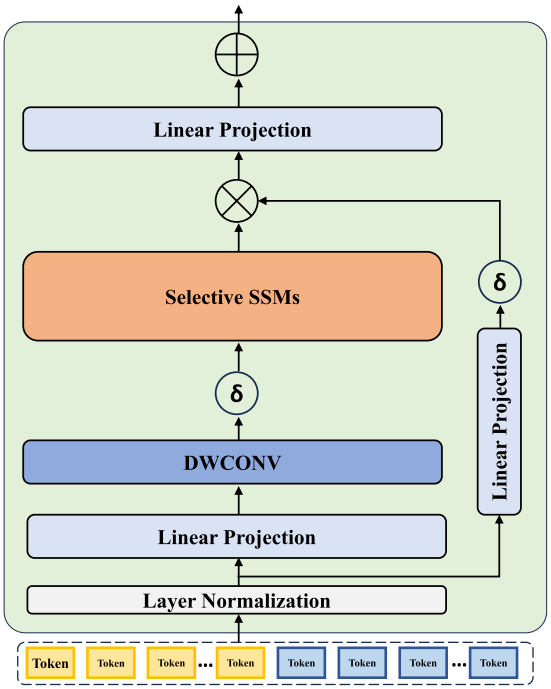


Fig. 1. Structure of the Mamba block, where δ represents the sigmoid linear unit function and DWConv represents a depth-wise convolution.

importance-aware ordering strategy to refine group embeddings and improve global feature aggregation, establishing a robust hybrid framework.

In these works, the serialization of point clouds is pivotal to the performance of Mamba. However, the impact of serialization strategies on Mamba's effectiveness and their underlying mechanisms remains underexplored. To bridge this gap, this article systematically compares and analyzes existing point cloud serialization techniques, introducing reliable indicators to evaluate serialization quality. Guided by these indicators, we proposed a novel serialization method that demonstrates superior performance on benchmark LiDAR datasets, named GSF serialization.

III. Method

This section first details the design of the Mamba block. Second, the baseline framework, PointMamba [6], is presented. Third, we investigated seven existing token serialization methods in Mamba-based point cloud processing networks and introduced their details. Fourth, the indicators for evaluating serialization quality are designed. Finally, a novel token serialization method (GSF) is proposed for Mamba performance improvement.

A. Mamba Block

As illustrated in Fig. 1, the Mamba block receives serialized tokens as input. It is built based on the S4 model [37], which is presented in detail as follows.

1) Structured State-Space Sequence Model (S4): The S4 architecture was recently developed based on SSMs [37]. Continuous-time SSMs are designed to map a sequence $x(t) \in \mathbb{R}$ to $y(t) \in \mathbb{R}$ by an implicit latent state $h(t) \in \mathbb{R}^N$. The SSM sequence-to-sequence transformation is defined as follows:

$$\begin{aligned} \acute{h}(t) &= Ah(t) + Bx(t) \\ y(t) &= Ch(t) \end{aligned} \tag{1}$$

where $A \in \mathbb{R}^{N\times N}$, $B \in \mathbb{R}^{N\times 1}$, and $C \in \mathbb{R}^{1\times N}$. Furthermore, to integrate SSMs into deep learning models, the parameters (A, B) in continuous-time SSMs are discretized into $(\bar{A}, \bar{B})$ by the zero-order hold (ZOH) discretization rule as follows:

$$\begin{aligned} \bar{A} &= \exp(\Delta A) \\ \bar{B} &= (\Delta A)^{-1}\left(\exp(\Delta A) - I\right) \cdot \Delta B \end{aligned} \tag{2}$$

where Δ is the time-scale parameter for discretization. Therefore, the new parameters $(\Delta, \bar{A}, \bar{B}, C$, and $D)$ define the S4 model, achieving discretized sequence transformation from $x(t)$ to $y(t)$, which can be expressed as follows:

$$\begin{aligned} h_t &= \bar{A}h_{t-1} + \bar{B}x_t \\ y_t &= Ch_t + Dx_t \end{aligned} \tag{3}$$

where $D \in \mathbb{R}^{1\times N}$ represents a residual connection.

2) Selective SSM (S6): Since the parameters in SSMs are fixed across the entire sequence processing, which is also called the linear time-invariant (LTI) property, the content-based reasoning capability of SSMs is limited. To tackle this limitation, Gu and Dao [1] proposed a novel parameterization method for SSMs. It enables dynamic adjustment of parameters $(\Delta, B$, and $C)$ according to the input data, allowing the model to dynamically transfer information across the sequence length axis. Due to the recurrence-based computation approach in S6, it requires the sequence data as input. In addition, considering the sparsity and uneven spatial distribution of point clouds, a depth-wise convolution (DWConv) [38] is introduced into the Mamba block for local feature enhancement, as shown in Fig. 1.

B. PointMamba Network

PointMamba offers a pure Mamba baseline for point cloud processing, with a plain architecture. As shown in Fig. 2, given the raw input $\mathbb{P} = \{p_i\}_{i\in N} \in R^{N\times C}$, where C is the feature dimension of input points, PointMamba first generates the sampling token set $\mathbb{S} = \{s_i\}_{i\in M} \in R^{M\times C}$ by the farthest point sampling (FPS) approach. Second, a Bi-Scanning strategy is proposed to achieve point cloud serialization, which includes a forward scanning path and the corresponding transposed path (also called the backward path). The Bi-Scanning strategy enables tokens to aggregate information from multiple directions, capturing intricate spatial relationships and providing richer global context for point cloud representation. Unlike Transformers, which use parallel attention for global dependencies, Mamba relies on sequential processing, making it sensitive to input order. Single-directional scanning struggles with distant token interaction, but Bi-Scanning addresses this by processing the point cloud bidirectionally, enlarging global receptive fields. Various specific point cloud serialization methods are detailed in Section III-C. By applying the Bi-Scanning strategy to $\mathbb{S}$, two different token serializations are

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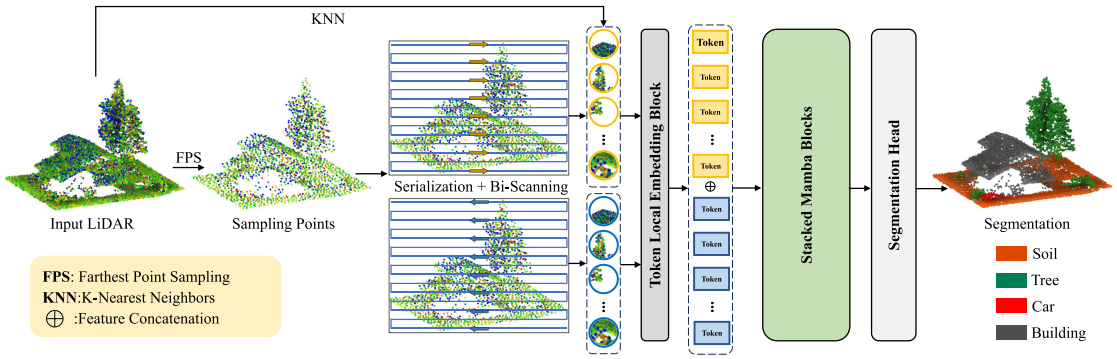


Fig. 2. Overall framework of PointMamba [6], illustrated with Bi-Scanning.

obtained, denoted as p_h and $p_{h'}$, where p_h represents the forward path and $p_{h'}$ is the corresponding backward path.

Given the two serialized token sets p_h and $p_{h'}$, a k-nearest neighbor (kNN) searching algorithm is utilized for local embedding for each sampling token. Specifically, kNN constructs a local neighborhood for each sampling token, followed by aggregating local information using a lightweight PointNet in the neighborhood. As such, two serialized token sets with local embedding, $E_h \in R^{M\times D}$ and $E_{h'} \in R^{M\times D}$, are generated, where D is the feature dimension after aggregation.

E_h and $E_{h'}$ are concatenated together as the input to stacked Mamba blocks described in Section III-A for further feature extraction. Finally, a segmentation head consisting of two fully connected layers with batch normalization and RELU activation is applied for the final prediction results.

C. Existing Point Cloud Serialization

We investigated seven existing point cloud serialization methods. Based on the feature types targeted by the involved serialization methods, they can be categorized into *geometric-based serialization* and *semantic-based serialization*. The geometric-based serialization method sorts tokens based on their spatial or geometric properties, which relies on the inherent spatial distribution of the tokens. It includes Hilbert space-filling curve, Z-order space-filling curve, radial sorting, lexicographic coordinate sorting, and ordering points to identify the clustering structure (OPTICS)-based ordering.[1] The semantic-based serialization method sorts tokens using learned deep features extracted by neural networks. DynamicViT [39] proposed a decision score calculation method based on deep semantic features for token importance assessment. Importance-aware ordering proposed in [5] also adopted the similar strategy for importance score prediction. In addition, we also introduced random sorting as a baseline reference to evaluate the impact of different serialization methods on Mamba's performance. The investigated serialization methods are detailed as follows.

1) Hilbert Space-Filling Curve: The Hilbert space-filling curve is a continuous fractal curve that passes through every point in a 3-D space within a predefined grid or cube. It recursively divides the space into smaller subcubes, ensuring that points that are close in 3-D space are also close in the 1-D ordering. The Hilbert curve is usually utilized to map multidimensional data to 1-D space while preserving spatial locality.

2) Z-Order Space-Filling Curve: The Z-order space-filling curve (also known as a Morton curve) is a recursive fractal curve that maps multidimensional data (e.g., 3-D points) into 1-D space. It is constructed by interleaving the binary representations of the coordinates of points in the 3-D space to create a single scalar value, known as the Z-order value or Morton code. Points are then sorted based on their Morton codes. The Z-order curve provides a means of sorting points by mapping their coordinates into a single dimension while preserving spatial locality to some extent.

3) Radial Sorting: Radial sorting is a method for ordering points in a 3-D point cloud based on their radial distances relative to a reference point (the geometric center point of the target point cloud in our experiments).

4) Lexicographic Coordinate Sorting: Lexicographic coordinate sorting is a straightforward method for ordering points in a 3-D point cloud based on their Cartesian coordinates (x, y, z) in a hierarchical, dictionary-like manner. The sorting process compares points coordinate by coordinate in a prioritized order, typically starting with the x-coordinate, then y, and finally z. For example, if two or more points have the same x-value, they are then sorted by the secondary coordinate (y). If the secondary coordinates are also identical, then the tertiary coordinate (z) is used for ordering. The sorted list of points follows the sequence defined by the lexicographic hierarchy.

5) OPTICS-Based Ordering: OPTICS-based ordering is derived from the OPTICS algorithm, which is commonly used for density-based clustering. OPTICS-based ordering focuses on arranging the points in a 3-D point cloud based on their core distances and reachability distances (assuming that all input points are core points for calculation simplification), producing a density-aware sequence. The result of this process is a sequence of points, where points that are close together in high-density areas are placed next to each other in order. In other words, it prioritizes arranging points in tightly packed regions first, while points in sparse areas or isolated points come later in the sequence.

6) Semantic-Based Ordering: The semantic-based ordering is guided by trainable decision scores, enabling a more adaptive and semantic-aware arrangement of tokens. To achieve this, we adopted the decision score calculation method

[1]OPTICS stands for Ordering Points To Identify the Clustering Structure.

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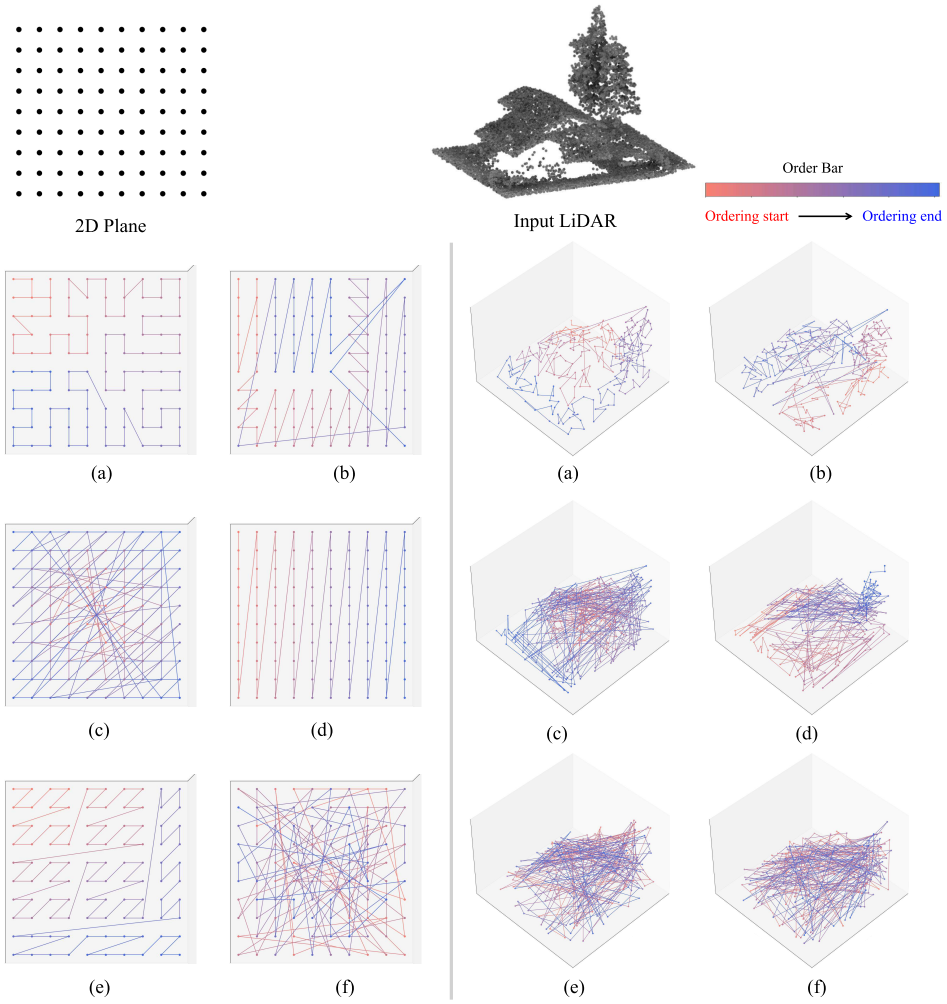


Fig. 3. Visualization of investigated serialization methods on uniformly distributed 2-D planar and complex 3-D LiDAR point clouds. A unified color bar (red to blue) accompanies all subplots, where red marks the beginning of the serialized sequence, while blue denotes its end. (a) Hilbert curve. (b) OPTICS. (c) Radial sorting. (d) Lexicographic. (e) Z-order curve. (f) Random.

proposed by DynamicViT [39]. Specifically, given the input tokens, an MLP is first applied to perform local embedding for each token individually. Then, average pooling is used to generate a global embedding that captures the overall representation of the token set. These local and global features are concatenated to form a global–local (GLocal) feature for each token, which integrates both local and global information. Finally, another MLP layer computes a decision score for each token, which serves as the basis for the ordering.

7) Random Ordering: Random sorting is introduced as a baseline reference to evaluate the impact of the aforementioned serialization methods on Mamba's performance. It assigns a random order to input tokens, without considering their spatial or geometric properties. In this process, all tokens are treated equally, ensuring a bias-free ordering where no point is prioritized over others.

To clearly illustrate the visualization results and differences among various serialization methods, Fig. 3 showcases the results of each method applied to a uniformly distributed 2-D planar point cloud and a LiDAR point cloud (downsampled from the DALES dataset [40]). From the results, Hilbert and OPTICS algorithms effectively preserve local geometric details in both 2-D planar point clouds and complex LiDAR point clouds. In contrast, although algorithms such as Lexicographic coordinate sorting and Z-order curve produce satisfactory sorting results on planar point clouds, they get chaotic ordering results on complex LiDAR point clouds, which tend to hinder their performance in LiDAR point cloud processing and analysis. This observation is further validated in subsequent experiments.

D. Serialization Indicators

After comparing and analyzing the investigated serialization methods, we further explored their inherent factors contributing to Mamba's performance. Different from NLP, point cloud data have its own topological geometric relationships, which need to be preserved during the serialization process. In addition, preserving the semantic homogeneity between adjacent points during serialization is also a potential indicator for evaluating serialization quality. Based on the aforementioned key observations, we proposed two kinds of evaluation indicators for serialization quality: spatial proximity and semantic proximity of serialized point clouds.

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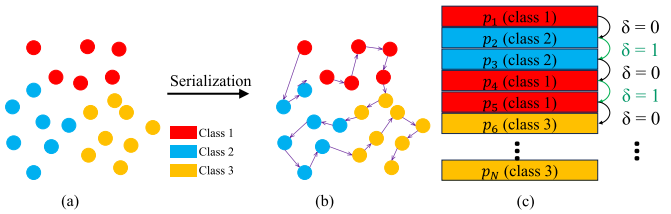


Fig. 4. Illustration of the semantic proximity definition. (a) Input points. (b) Serialized points. (c) Semantic proximity.

In this section, we express a point as $p_i = (\vec{x_i}, \vec{f_i})$ with a semantic label L_i, where $\vec{x_i} = \{X_i, Y_i, Z_i\}$ is a 3-D geometric component, and $\vec{f_i}$ is a D-dimensional semantic feature component.

Spatial Proximity: It refers to whether adjacent pairs of points p_i and p_j in the serialized point cloud remain close in the original 3-D space. This can be quantitatively measured by the normalized sum of Euclidean distance between adjacent pairs of points given by the following equation:

$$\text{Spatial proximity} = \frac{1}{N-1}\sum_{i=1}^{N-1} d\left(\vec{x_i}, \vec{x}_{i+1}\right) \tag{4}$$

where

$$d\left(\vec{x_i}, \vec{x_j}\right) = \sqrt{\left(X_i - X_j\right)^2 + \left(Y_i - Y_j\right)^2 + \left(Z_i - Z_j\right)^2}. \tag{5}$$

We traverse all adjacent point pairs in the serialized point cloud and sum their geometric distances. The smaller the distance, the higher the spatial proximity.

Semantic Proximity: On the other hand, it quantifies the degree to which consecutive points in the serialized point cloud belong to the same semantic category. In point cloud segmentation, preserving semantic continuity in the serialized order is critical, as it helps the model capture both local context and long-range dependencies. As shown in Fig. 4, it is defined as the proportion of adjacent point pairs in the serialized sequence that share the same semantic label relative to the total number of *adjacent point pairs*. It is given by the following equation:

$$\text{Semantic proximity} = \frac{1}{N-1}\sum_{i=1}^{N-1} \delta\left(L_i, L_{i+1}\right) \tag{6}$$

where L_i represents the class label of point p_i and δ is the Kronecker delta function given by the following equation:

$$\delta(a,b) = \begin{cases} 1, & \text{if } a = b \\ 0, & \text{if } a \neq b. \end{cases} \tag{7}$$

A higher semantic proximity value indicates that the serialization method has effectively grouped points of the same semantic category together. This preservation of semantic homogeneity not only reflects a faithful representation of the underlying scene but also benefits downstream processing, particularly in Mamba-based architectures where the order of tokens is crucial for leveraging sequential dependencies.

In contrast, a lower semantic proximity suggests that points from different semantic classes are interleaved in the serialized sequence, potentially disrupting semantic coherence and degrading the model's segmentation performance. Therefore, semantic proximity serves as a vital indicator when evaluating and designing point cloud serialization methods, especially when used in conjunction with spatial proximity metrics to provide a comprehensive measure of serialization quality.

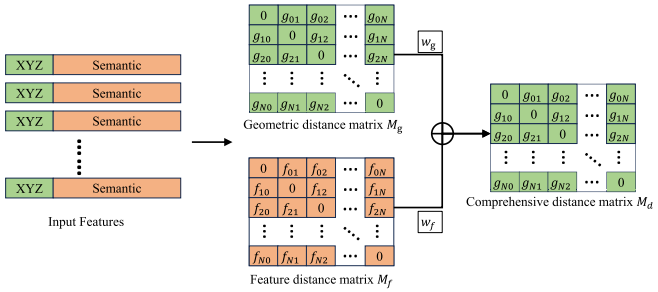


Fig. 5. Generation of the comprehensive distance matrix M_d.

E. Serialization Improvement

The investigated serialization methods focus only on spatial structure or semantic information (semantic-based ordering) of the target point cloud, which is not comprehensive and limits their performance on Mamba. Guided by the proposed indicators for serialization quality evaluation, we proposed a novel point cloud serialization method, named GSF serialization, effectively incorporating both spatial and semantic proximity.

We explicitly integrate geometric and semantic features for point cloud serialization. As shown in Fig. 2, after FPS and kNN-based local feature aggregation, a token local embedding block is applied to refine feature representation. The features emitted by this block are used as the semantic embeddings in GSF, as they encode local aggregation features and are fine-grained enough to distinguish neighboring points. As shown in Fig. 5, given a point cloud with geometric coordinates and embedded features, we first construct a geometric distance matrix M_g of size $N \times N$, where each element represents the geometric distance between two corresponding points. Subsequently, we generate a feature distance matrix M_f of size $N \times N$ based on the distance between embedded features of the point cloud. This is given by the following equation:

$$\begin{aligned} g_{ij} &= d\left(\vec{x_i}, \vec{x_j}\right) \\ f_{ij} &= d\left(\vec{f_i}, \vec{f_j}\right) \end{aligned} \tag{8}$$

where g_{ij} and f_{ij} are the element at ith row and jth column in M_g and M_f, respectively. To account for both spatial and semantic proximity during serialization, a weighted average of the two matrices is computed to obtain a final comprehensive distance matrix M_d, which is expressed as follows:

$$M_d = w_g M_g + w_f M_f \tag{9}$$

where w_g and w_f represent the weights of M_g and M_f, respectively. $w_g = 2.0$ and $w_f = 1.0$ in our experiments. Using the matrix M_d, we apply the minimum spanning tree (MST) algorithm to derive the serialized point cloud sequence.

IV. Experiments and Results

This section first shows the implementation settings of the baseline model. Second, we introduced the LiDAR semantic

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TABLE I
COMPARISON OF THE AIRBORNE MS-LIDAR AND AERIAL DALES DATASETS

Aspect	MS-LiDAR	DALES
Collection Area	Canadian town	Urban and suburban areas in Canada
Resolution	$\sim$3.6 points/m^2	$\sim$50 points/m^2
Classes	6 (road, building, grass, tree, soil, powerline)	8 (ground, vegetation, buildings, cars, trucks, poles, power lines, fences)
Data Scale	13 areas, 15,000 m^2/area $\sim$10 million points total	40 tiles, 0.5 km^2/tile $\sim$500 million points total
LiDAR Type	Multispectral	Aerial
Point Attributes	XYZ, intensity (MIR, NIR, Green)	XYZ, intensity
Applications	Land cover classification, Urban planning, and Environmental monitoring	Urban modeling, Environmental monitoring

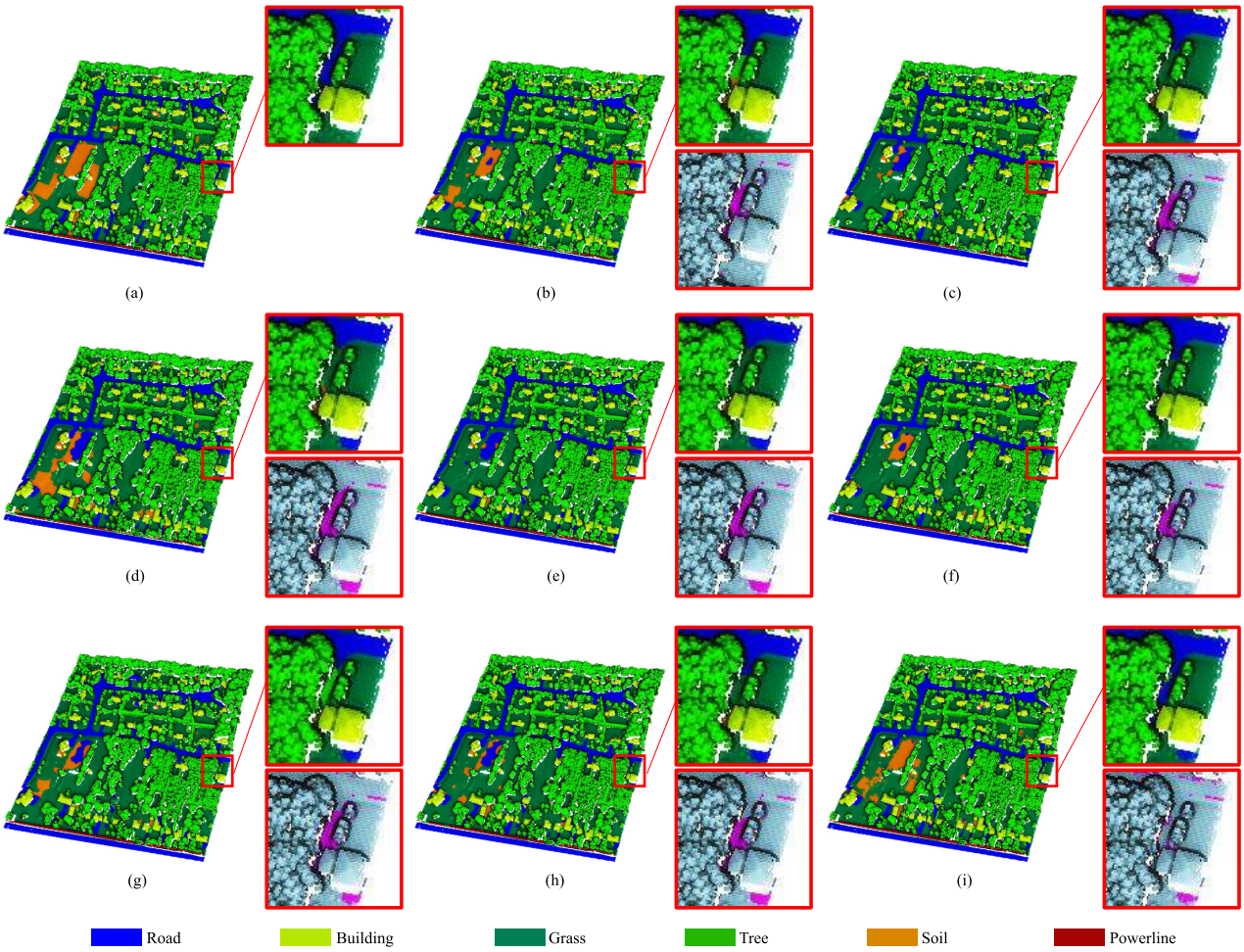


Fig. 6. Visualization comparison results of different serialization methods on the MS-LiDAR dataset. Both segmentation results and corresponding error maps of zoom-in areas are shown for clear visualization, where the pink points are misclassified points. (a) Ground truth. (b) PointMamba + random ordering. (c) PointMamba + Z-order curve. (d) PointMamba + semantic ordering. (e) PointMamba + raidal ordering. (f) PointMamba + lexicographic ordering. (g) PointMamba + OPTICS ordering. (h) PointMamba + Hilbert curve. (i) PointMamba + GSF (ours).

segmentation datasets used in our experiments, as well as the measurement metrics. Third, we present and analyze the comparison results of all involved point cloud serialization methods on the datasets.

A. Implementation Details

PointMamba [6], the baseline model employed in our experiments, was implemented in PyTorch and executed on NVIDIA Tesla V100 GPUs. The training process utilized the SGD optimizer, configured with a momentum of 0.9 and a weight decay of 0.0001. The learning rate was initialized at 0.01 and progressively adjusted at each epoch using a cosine annealing schedule. The model was trained for 200 epochs for each dataset application.

B. Datasets and Metrics

To fully evaluate the contributions of serialization methods to Mamba-based LiDAR point cloud processing, we utilized two challenging LiDAR datasets, covering airborne MS-LiDAR and aerial LiDAR data. These datasets, sourced from different regions, represent diverse point cloud categories and varying levels of scene complexity. Such diversity

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TABLE II
PERFORMANCE COMPARISON OF DIFFERENT METHODS ON THE AIRBORNE MS-LIDAR DATASET

Methods	Input Points	Average F_1 score (%)	mIoU (%)	OA (%)	Latency (ms)
Other Deep Learning Models					
PointNet++ [17]	4096	72.1	58.6	90.1	322.6
KPConv [41]	4096	76.8	64.4	92.2	83.7
DGCNN [20]	4096	71.6	51.0	91.4	86.2
RSCNN [42]	4096	73.9	56.1	91.0	158.7
GACNet [43]	4096	67.7	51.0	90.0	277.8
AGConv [44]	4096	76.9	71.2	93.3	312.5
SE-PointNet++ [45]	4096	75.9	60.2	91.2	-
FR-GCNet [46]	4096	78.6	65.8	93.6	-
ResMRGCN-28 [47]	4096	81.1	74.0	93.3	45.7
Transformer-based Models					
PCT [23]	4096	79.3	71.3	92.4	47.1
PointTransformer2 [24]	4096	80.5	73.6	93.1	285.7
Xiao et al. [48]	4096	83.3	79.3	94.0	-
PatchFormer [28]	4096	82.4	77.8	93.1	62.9
DCTNet [49]	4096	86.0	80.2	95.0	**23.3**
Mamba-based Models					
PointCloudMamba [4]	4096	86.1	80.5	94.8	52.7
PointMamba [6] + Random Ordering	4096	84.4	76.1	93.6	54.3
PointMamba [6] + Z-order Curve	4096	84.9	76.1	93.8	55.3
PointMamba [6] + Semantic Ordering	4096	84.5	76.1	93.9	54.3
PointMamba [6] + Radial Ordering	4096	85.1	76.3	93.9	54.3
PointMamba [6] + Lexicographic Ordering	4096	85.2	76.4	94.1	54.3
PointMamba [6] + OPTICS Ordering	4096	86.0	77.2	93.9	214.6
PointMamba [6] + Hilbert Curve (original PointMamba)	4096	86.3	77.6	94.2	58.3
PointMamba [6] + GSF (ours)	4096	**88.7**	**81.4**	**95.1**	54.6

TABLE III
PERFORMANCE COMPARISON OF DIFFERENT METHODS ON THE DALES DATASET

Methods	Input Points	mIoU (%)	OA (%)	Latency(ms)
Other Deep Learning Models				
PointNet++ [17]	8192	68.3	95.7	487.1
KPConv [41]	8192	72.4	96.9	125.3
DGCNN [20]	8192	66.4	96.1	136.2
PointCNN [18]	8192	58.4	97.2	-
SPG [50]	8192	60.6	95.5	-
ConvPoint [51]	8192	67.4	97.2	-
Transformer-based Models				
PCT [23]	8192	72.7	96.3	**75.8**
PointTransformer2 [24]	8192	74.9	97.1	468.4
SPT [52]	8192	**79.6**	**97.5**	-
SuperCluster [53]	8192	77.3	-	-
PReFormer [54]	8192	70.9	92.9	-
Mamba-based Models				
PointCloudMamba [4]	8192	75.7	97.0	77.5
PointMamba [6] + Random Ordering	8192	71.1	96.1	82.3
PointMamba [6] + Z-order Curve	8192	71.2	96.1	84.5
PointMamba [6] + Semantic Ordering	8192	71.6	96.3	82.4
PointMamba [6] + Radial Ordering	8192	71.6	96.2	82.4
PointMamba [6] + Lexicographic Ordering	8192	72.4	96.3	82.3
PointMamba [6] + OPTICS Ordering	8192	73.3	96.4	412.5
PointMamba [6] + Hilbert Curve (Original PointMamba)	8192	73.3	96.3	91.8
PointMamba [6] + GSF (ours)	8192	76.2	97.4	82.5

guarantees a comprehensive evaluation of model performance across a range of LiDAR application scenarios strengthening the reliability and consistency of the comparative results and eliminating experimental randomness.

1) Airborne MS-LiDAR Dataset: The airborne MS-LiDAR dataset was proposed in [46]. It contains 13 areas in total. Specifically, the training set contains No.1–10 areas, and the testing set contains No.11–13 areas. Each area covers over 15 000 m^2 with an average point density of about 3.6 points/m^2. Each point has six attributes: *XYZ*, MIR (1550-nm wavelength), NIR (1064-nm wavelength), and Green (532-nm wavelength). There are six categories in the datasets: road, building, grass, tree, soil, and powerline. The imbalanced class distribution makes the dataset more challenging, where the number of road points is more than 30× the number of powerline points. Similar data preprocessing methods to

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TABLE IV
RUNNING TIME (MS) COMPARISON OF DIFFERENT SERIALIZATION METHODS. A SINGLE SAMPLE OF EACH DATASET IS TAKEN AS INPUT

Methods	MS-LiDAR				DALES			
	Downsampling	Serialization	Inference	Total	Downsampling	Serialization	Inference	Total
Random Ordering	41.6	**0.4**	**12.3**	**54.3**	66.5	**0.6**	**15.2**	**82.3**
Z-order Curve	41.6	1.4	12.3	55.3	66.5	2.8	15.2	84.5
Semantic Ordering	41.6	0.4	12.7	54.3	66.5	0.6	15.9	82.4
Radial Ordering	41.6	0.4	12.3	54.3	66.5	0.7	15.2	82.4
Lexicographic Ordering	41.6	0.4	12.3	54.3	66.5	0.6	15.2	82.3
OPTICS Ordering	41.6	160.7	12.3	214.6	66.5	330.8	15.2	412.5
Hilbert Curve	41.6	4.4	12.3	58.3	66.5	10.1	15.2	91.8
GSF (ours)	41.6	0.7	13.0	54.6	66.5	0.8	16.0	82.5

[46] are applied in our experiments, ensuring a fair comparison. Each area was split into a series of local blocks by kNN searching. k is set to 4096 in our experiments. The mean intersection over union (mIoU), overall accuracy (OA), average $F1$ score, and latency[2] are used for performance evaluation.

2) DALES Dataset: The DALES dataset was presented in [40]. It is an extensive aerial LiDAR dataset comprising more than 500 million points distributed over ten square kilometers. The dataset is classified into eight distinct object categories: ground, vegetation, cars, trucks, powerlines, fences, poles, and buildings. Point clouds obtained from Aerial Laser Scanners present unique challenges and offer novel applications. The DALES dataset was split into 40 areas with the elaborating class annotation, each spanning 0.5 km^2 and containing 12 million points. Each input point has four attributes (XYZ + Intensity). For a fair comparison, we subsampled each area using a 10-cm grid, followed by dividing it into a series of 20$\times$ 20 m blocks as training/testing samples, where each of them contained 8192 points after sampling. Similarly, the mIoU, OA, and latency are used for evaluation.[3]

A brief comparison of the aforementioned LiDAR datasets is shown in Table I, highlighting the diversity of LiDAR point cloud data used in this study.

C. Performance Comparison

We tested all eight serialization methods (including existing and newly proposed GSF methods) introduced in Section III on these two datasets and compared their performance with existing point cloud segmentation methods.

As shown in Tables II and III, we present the comparative performance of various algorithms on the MS-LiDAR and DALES datasets. On the MS-LiDAR dataset, the proposed PointMamba+GSF (ours) method achieves the best segmentation accuracy among all compared techniques, outperforming existing serialization strategies. Specifically, it records an average $F1$ score of 88.7%, an mIoU of 81.4%, and an OA of 95.1%. On the DALES dataset, PointMamba+GSF (ours) attains an OA of 97.4% and an mIoU of 76.2%, second only to SPT [52], which reports an mIoU of 79.6% and outperforms other segmentation models. By comparing the investigated serialization methods, we notice that they exhibit consistent trends across the two datasets, which eliminates experimental randomness and ensures reliability. For example, GSF achieves the best results across all comparisons, while the accuracies of Z-order curve and random ordering are similar and are the lowest. Figs. 6 and 7 show the visualization results of segmentation comparison among all involved serialization methods, on both MS-LiDAR and DALES datasets. These comparative results lay the foundation for the proposed serialization indicators.

As shown in Table IV, the choice of serialization method has a minimal impact on runtime efficiency. For most methods, serialization contributes only 1~2 ms to the total runtime, with the exception of OPTICS ordering, which incurs significantly higher latency (e.g., 160.7 ms on MS-LiDAR). Furthermore, the efficiency analysis reveals that the inference stage accounts for approximately 22% of the total runtime per sample on both datasets, whereas the downsampling procedure using FPS dominates the runtime, consuming over 70% of the total time. These results indicate that the impact of serialization is primarily on improving accuracy rather than efficiency, as the serialization time is negligible compared to other stages.

The two benchmarks employed in this study are quite different in point density, scene complexity, and class distribution. However, GSF still attains outperforming results on both of them, as well as maintaining consistently excellent segmentation results for key terrain-related classes such as forest and ground. These robust categories could contribute to practical downstream tasks, such as large-area biomass estimation and accurate digital terrain model (DTM) generation. The cross-dataset consistency stems from the geometry–semantic fusion of GSF. Specifically, by jointly exploiting geometric and semantic context, the GSF-based Mamba model adapts to point clouds of widely varying densities while faithfully preserving the underlying 3-D geometry and semantic structures. Although this study has not yet been evaluated on autonomous-driving datasets, the strong performance of the GSF-enhanced Mamba model on the highly sparse MS-LiDAR scenes demonstrates its superiority in processing low-density point clouds and therefore highlights its considerable potential for extension to the autonomous-driving domain. Real-time deployment, however, imposes strict latency constraints. Therefore, further optimizing the overall pipeline efficiency is a critical prerequisite for transferring the model to autonomous-driving scenarios.

[2]The latency refers to the time (ms) required for a model to process the input data. For a fair comparison, the indicator of latency is measured on an NVIDIA GeForce RTX 2080 GPU, and the batch size of the input points is set to 1.

[3]We notice that the vast majority of the algorithms compared on the DALES dataset do not report $F1$ scores. Accordingly, the $F1$ score has been excluded from the evaluation metrics for this dataset.

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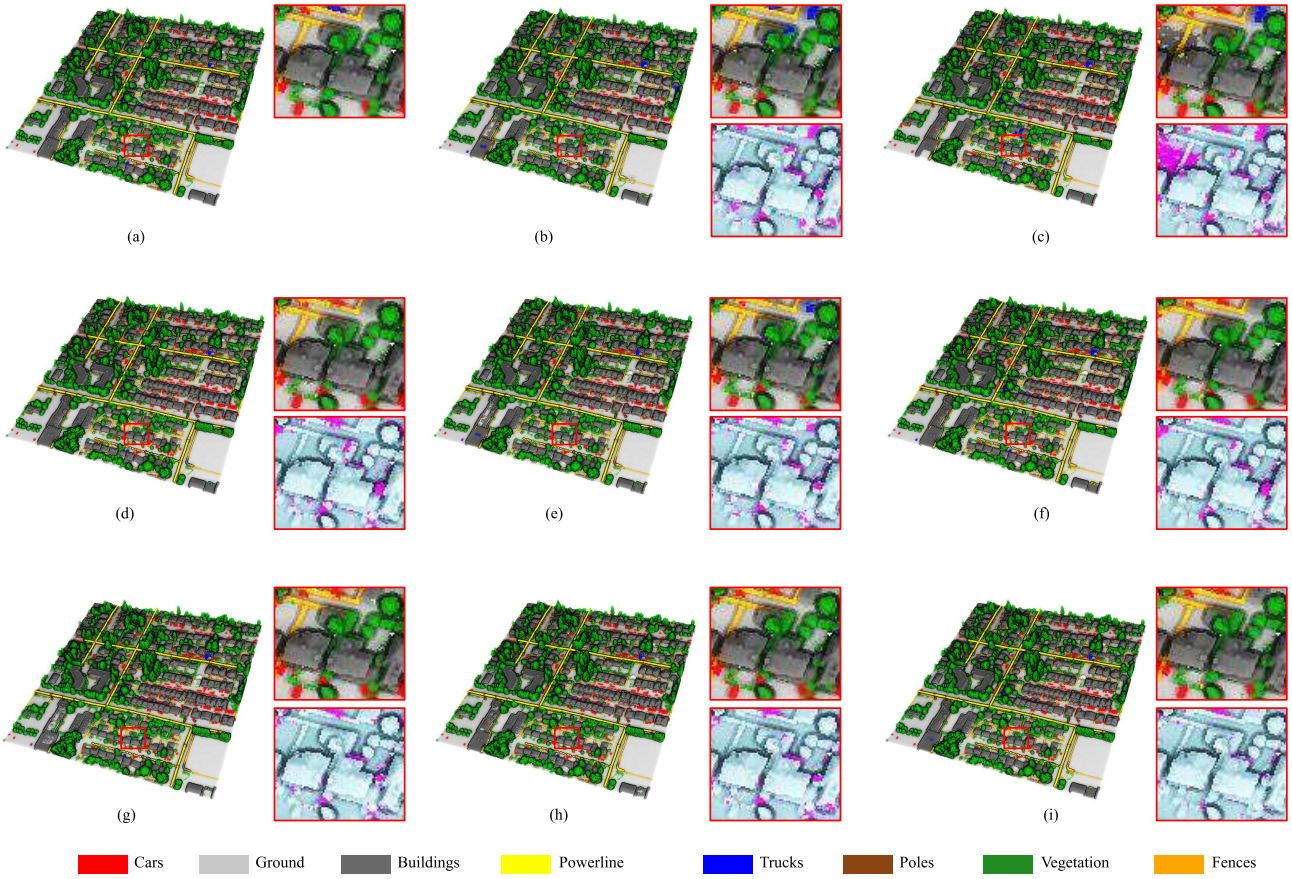


Fig. 7. Visualization comparison results of different serialization methods on the DALES dataset. Both segmentation results and corresponding error maps of zoomed-in areas are shown for clear visualization, where the pink points are misclassified points. (a) Ground truth. (b) PointMamba + random ordering. (c) PointMamba + Z-order curve. (d) PointMamba + semantic ordering. (e) PointMamba + raidal ordering. (f) PointMamba + lexicographic ordering. (g) PointMamba + OPTICS ordering. (h) PointMamba + Hilbert curve. (i) PointMamba + GSF (ours).

D. Indicator Evaluation and Comparison

We applied the proposed indicators in Section III to the investigated serialization methods, to verify their effectiveness in serialization quality evaluation. The comparison results measured by the proposed indicators are shown in Fig. 8. For clear presentation, the comparison results are shown in the form of both tables and curves. By comparing the accuracy curves and corresponding indicator curves of the serialization methods across the two datasets in Fig. 8, it is evident that serialization methods with higher accuracy are associated with higher spatial proximity (smaller pair distances) and higher semantic proximity (greater proportion of semantical homogeneous point pairs). For example, the proposed GSF serialization method, by incorporating both spatial and semantic information for token serialization, achieves the best performance across all two datasets. In addition, it yields the best spatial and semantic proximity.

Compared with the Hilbert curve, the Z-order curve has relatively weak performance in both LiDAR datasets with corresponding worse indicators. The main reason why the Hilbert curve outperforms Z-order in LiDAR point cloud serialization is discussed as follows. As shown in Fig. 3, the recursive construction of the Hilbert curve ensures adjacent points in the 3-D space remain close in the serialized sequence, making it highly effective in retaining both geometric and semantic structures of dense and complex LiDAR scenes. In contrast, Z-order's reliance on simple Morton coding often results in large jumps in point sequences, disrupting local relationships and leading to poor spatial and semantic proximity. This limitation is particularly pronounced in LiDAR data, where intricate geometries and uneven distributions are common, causing Z-order's performance to approach that of random sorting.

Therefore, the quality of serialization is generally positively correlated with the proposed spatial and semantic proximity indicators, as demonstrated consistently across all two datasets. These results validate the effectiveness and superiority of the proposed GSF serialization method while supporting the rationality of the evaluation indicators for point cloud serialization.

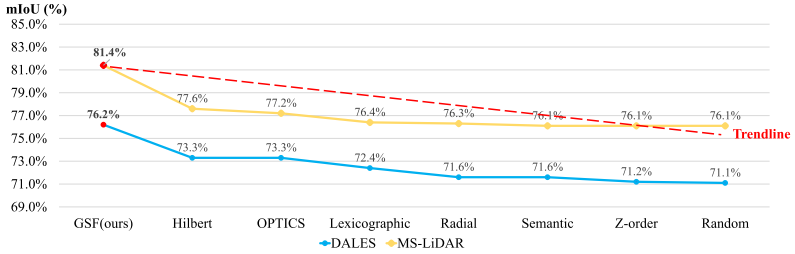
E. Ablation Study

To further evaluate the effectiveness and robustness of GSF, a series of ablation experiments was conducted.

1) Fusion Weights: As shown in (9), two fusion weights (w_g and w_f) are used to balance the model's attention to spatial and semantic relationships among input points, which were empirically tuned in our work. As shown in Table V, a series of different fusion weight combinations were explored to achieve the best performance. First, we separately set the weights w_g and w_f to 0.0, i.e., removing the spatial and semantic relationships between points, respectively, during the serialization

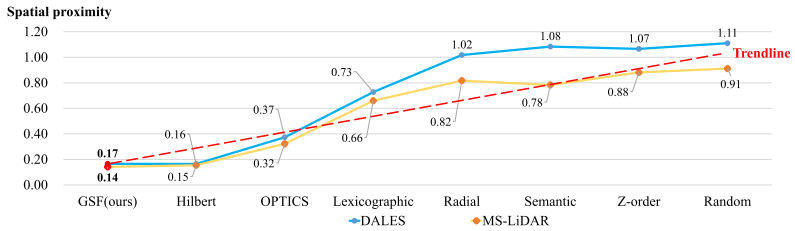
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Serialization	mIoU (%)	
	MS-LiDAR	DALES
Random Ordering	76.1	71.1
Z-order Curve	76.1	71.2
Semantic Ordering	76.1	71.6
Radial Ordering	76.3	71.6
Lexicographic Ordering	76.4	72.4
OPTICS Ordering	77.2	73.3
Hilbert Curve	77.6	73.3
GSF (ours)	**81.4**	**76.2**



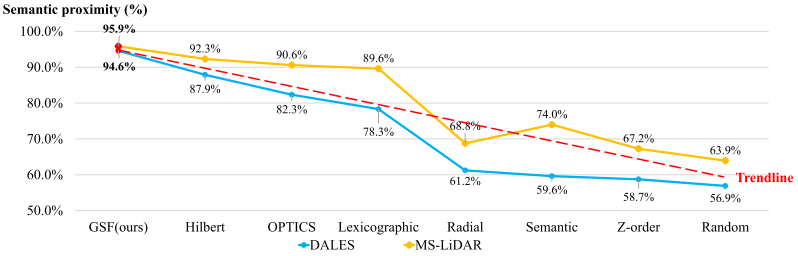
(a)

Serialization	Spatial proximity	
	MS-LiDAR	DALES
Random Ordering	0.91	1,11
Z-order Curve	0.88	1.07
Semantic Ordering	0.78	1.08
Radial Ordering	0.82	1.02
Lexicographic Ordering	0.66	0.73
OPTICS Ordering	0.32	0.37
Hilbert Curve	0.15	**0.16**
GSF (ours)	**0.14**	0.17



(b)

Serialization	Semantic proximity (%)	
	MS-LiDAR	DALES
Random Ordering	63.9	56.9
Z-order Curve	67.2	58.7
Semantic Ordering	74.0	59.6
Radial Ordering	68.8	61.2
Lexicographic Ordering	89.6	78.3
OPTICS Ordering	90.6	82.3
Hilbert Curve	92.3	87.9
GSF (ours)	**95.9**	**94.6**



(c)

Fig. 8. Comparison results of the proposed serialization method across two LiDAR datasets, measured by mIoU, spatial proximity, and semantic proximity. (a) Comparison results of the proposed serialization method measured by mIoU across two LiDAR datasets. (b) Comparison results of the proposed serialization method measured by the indicator of spatial proximity across two LiDAR datasets. (c) Comparison results of the proposed serialization method measured by the indicator of semantic proximity across two LiDAR datasets.

TABLE V

ABLATION RESULTS OF MODEL PERFORMANCE UNDER DIFFERENT FUSION WEIGHT COMBINATIONS, MEASURED BY MIOU (%)

Fusion weights		MS-LiDAR	DALES
w_g	w_f		
1.0	0.0	77.5	73.2
0.0	1.0	77.2	72.7
1.0	1.0	80.1	75.1
1.0	2.0	79.7	74.4
1.0	3.0	79.9	74.4
2.0	**1.0**	**81.4**	**76.2**
3.0	1.0	80.5	75.7

process. We observed a performance drop in both cases, indicating that both spatial and semantic relationships contribute to point cloud serialization. Second, from the results, the combination of both spatial and semantic relationships among points results in a significant improvement to the model performance. The GSF method consistently gets the best results on both datasets when $w_g = 2.0$ and $w_f = 1.0$. This result suggests that the combination of spatial and semantic relationships plays a critical role in point sequence construction. By assigning a relatively higher weight to the spatial term ($w_g = 2.0$), the model is encouraged to prioritize the preservation of geometric proximity and structural continuity during serialization, which is particularly important for capturing local context in point cloud data. In addition, the semantic term allows the model to consider semantic feature similarity, helping to maintain semantic coherence within the sequence. The combination of these two factors leads to more informative and structured sequences, thereby improving downstream performance.

2) Sensitivity to Noise: To evaluate the robustness of the proposed GSF model against noise, we conducted a series of ablation experiments by injecting varying levels of zero-mean Gaussian noise into the point cloud data. Specifically, we applied noise to the position of each point as Gaussian distributed random displacements, with standard deviations corresponding to 0.05%, 0.1%, 0.5%, 1%, 2%, and 5% of the spatial range of the point cloud. The noise was applied to the 3-D coordinates (*XYZ*) of all points in both datasets, simulating real-world sensor perturbations or registration inaccuracies.

Specifically, we implemented two training strategies. The first strategy involves adding the same level of noise to both the training and testing datasets, ensuring that the training data are augmented with noise. The second strategy trains the model on a clean dataset without additional noise, while noise of varying intensities is introduced only to the testing data for performance evaluation. These two strategies allow us to systematically assess the impact of noise on the model's generalization ability and robustness. As shown in Fig. 9, with noise-augmented training data, the model maintains stable performance under varying levels of noise ($\leq$ 2% Gaussian noise). However, a relatively noticeable performance degradation occurs when the data are heavily corrupted by noise, particularly under severe conditions such as 5% Gaussian

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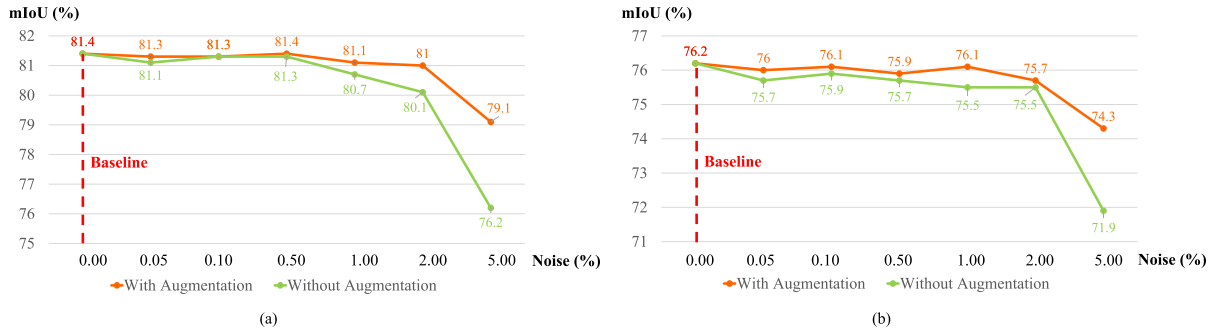


Fig. 9. Ablation results of the model's robustness to different levels of noise on the MS-LiDAR and DALES datasets, measured by mIoU. (a) Ablation results of the model's robustness to different levels of noise on the MS-LiDAR dataset. (b) Ablation results of the model's robustness to different levels of noise on the DALES dataset.

noise. In contrast, in the case that the training data are not augmented with noise, the model's robustness to noise decreases significantly. It can still maintain relatively stable performance under mild noise conditions (less than 1% Gaussian noise). However, when the noise level exceeds 2%, the performance degrades noticeably. This indicates that the model has limited robustness to noise without noise-augmented training data. We attribute this to the fact that the spatial relationship used in MST-based point cloud serialization is inherently sensitive to noise. When the model is trained with noise-augmented data, it learns to extract more robust semantic features from noisy input, which mitigates the impact of noise on the MST-based serialization process. In contrast, without noise augmentation during training, the model generalizes poorly to noisy test data and is therefore more susceptible to performance degradation caused by noise interference.

V. Discussion

The importance of point cloud serialization in Mamba-based models stems from Mamba's inherent recurrence mechanism, which requires sequenced data as input. The token order directly determines how information is propagated along the sequence, thus significantly affecting the model's ability to capture spatial and semantic dependencies. In contrast, Transformer models adopt a parallel attention mechanism and are fundamentally permutation-invariant. The self-attention operation computes pairwise interactions among all tokens irrespective of their input order. Moreover, relative positional encoding explicitly integrates spatial information into the attention map by modeling pairwise geometric relationships, enabling point cloud Transformers to effectively capture spatial structures while remaining insensitive to input token order. The experimental results demonstrate that the proposed GSF serialization significantly improves the performance of Mamba-based point cloud segmentation across both the MS-LiDAR and DALES datasets. By effectively integrating spatial and semantic proximity, GSF outperforms traditional geometric-based (e.g., Hilbert and OPTICS) serialization methods, achieving better accuracies with minimal additional computational cost. This confirms that enhancing serialization quality is crucial for maximizing the potential of Mamba in point cloud processing.

Despite its strong performance, the GSF algorithm still has limitations and room for further exploration. First, the noise sensitivity analysis reveals that MST-based serialization is inherently sensitive to noise, leading to noticeable performance degradation when the input data is corrupted, especially without noise-augmented training. This limitation stems from the fact that noise can perturb point coordinates, which in turn affects the computation of distance matrices and alters the structure of the MST, thereby disrupting the spatial continuity of the serialized sequence. Second, GSF relies on the assumption of global graph connectivity, which can be problematic in the presence of class imbalance. In scenes where small objects (e.g., powerlines and cars) are sparsely distributed relative to large objects (e.g., buildings and vegetation), it tends to generate long edges that span between disparate classes, undermining semantic continuity. Addressing these challenges is a key direction for future research. Inspired by Superpoint Transformer [52], one promising solution is to adopt a hierarchical superpoint-graph approach, where the point cloud is first clustered into semantically homogeneous superpoints at multiple scales. The GSF+Mamba method can then be applied within each superpoint cluster, ensuring that serialization operates in a context of local semantic consistency.

In addition, with the widespread application of large-scale LiDAR point clouds, transfer or weakly supervised learning is increasingly crucial in remote sensing applications, as they alleviate exhaustive manual labeling and accelerate city- or country-level LiDAR data processing. Owing to the joint exploitation of geometric and semantic context, the proposed GSF method is, in principle, more adaptable to domain shifts than purely geometric curves. It preserves local neighborhood structure information via high spatial proximity while introducing further constraint of semantic proximity in data sequence, thereby providing an ordering that remains meaningful across datasets with different data distributions. We believe that integrating GSF with a domain-adaptive Mamba backbone will facilitate the learning of generalizable features for robust transfer learning or weakly supervised learning.

VI. Conclusion

This study highlights the crucial impact of token serialization on Mamba-based point cloud processing. The proposed GSF method significantly improves model accuracy by enhancing spatial and semantic proximity within token

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sequences. Despite its effectiveness, GSF still faces limitations, particularly its sensitivity to noise and challenges in handling class imbalance due to the reliance on global graph connectivity. To address these issues, our future work will focus on improving the MST-based ordering strategy in GSF to develop more robust and adaptive serialization methods, as well as exploring hierarchical superpoint-based Mamba frameworks to further enhance robustness and scalability in complex point cloud scenes.

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Dening Lu received the B.Sc. and M.Sc. degrees in electrical engineering from Nanjing University of Aeronautics and Astronautics (NUAA), Nanjing, China, in 2018 and 2021, respectively. He is currently pursuing the Ph.D. degree in systems design engineering with the Geospatial Intelligence and Mapping Laboratory, University of Waterloo, Waterloo, ON, Canada.

He has published articles in IEEE TRANSACTIONS ON INSTRUMENTATION AND MEASUREMENT, IEEE TRANSACTIONS ON INTELLIGENT TRANSPORTATION SYSTEMS, IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING, *International Journal of Applied Earth Observation and Geoinformation*, and ICCV. His research interests include 3-D point cloud processing and deep learning.



Kyle Gao (Member, IEEE) received the bachelor's degree in mathematics from the University of Waterloo, Waterloo, ON, Canada, in 2016, and the master's degree in physics from the University of Victoria, Victoria, BC, Canada, in 2020. He is currently pursuing the Ph.D. degree in systems design engineering with the Geospatial Intelligence and Mapping Laboratory, University of Waterloo.

He has published articles in IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING, *International Journal of Applied Earth Observation and Geoinformation*, *Geomatica*, and IEEE TRANSACTIONS ON INTELLIGENT TRANSPORTATION SYSTEMS. His research interests include computer vision and deep learning.



Jonathan Li (Fellow, IEEE) received the Ph.D. degree in geomatics engineering from the University of Cape Town, Cape Town, South Africa, in 2000.

He is currently a Professor of geomatics and systems design engineering at the University of Waterloo, Waterloo, ON, Canada. He has supervised more than 200 master's/Ph.D. students and postdoctoral fellows/visiting scholars to completion. He has co-authored almost 600 publications, more than 200 of which were published in top remote sensing journals, including *Remote Sensing of Environment*, *ISPRS Journal of Photogrammetry and Remote Sensing*, and *International Journal of Applied Earth Observation and Geoinformation* (JAG), and IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING. He has also published articles in flagship conferences in computer vision and AI, including NeurIPS, CVPR, AAAI, and IJCAI. His main research interests include AI-based information extraction from Earth observation images and LiDAR point clouds, pointgrammetry and remote sensing, GeoAI, and 3-D vision for digital twin cities and autonomous driving.

Dr. Li is a fellow of Canadian Academy of Engineering, the Royal Society of Canada (Academy of Science), and the Engineering Institute of Canada. He is the elected President of Canadian Institute of Geomatics (CIG) and an elected Advisor of the Committee of Experts on United Nations Global Geospatial Information Management (UN-GGIM) Academic Network. He is the Editor-in-Chief of JAG and an Associate Editor of IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING.



Dedong Zhang (Member, IEEE) received the B.Sc. and M.Sc. degrees in electrical and computer engineering from Baylor University, Waco, TX, USA, in 2016 and 2018, respectively, and the Ph.D. degree in systems design engineering from the University of Waterloo, Waterloo, ON, Canada, in 2023.

He is currently a Post-Doctoral Researcher with the Department of Geography and Environmental Management, University of Waterloo. He specializes in optical, IMU, and LiDAR remote sensing; machine learning; artificial intelligence; and simultaneous localization and mapping. His research interests include computer vision, localization and mapping, and multisensor fusion.



Linlin Xu (Member, IEEE) received the B.Eng. and M.Sc. degrees in geomatics engineering from China University of Geosciences, Beijing, China, in 2007 and 2010, respectively, and the Ph.D. degree in geography from the University of Waterloo, Waterloo, ON, Canada, in 2014.

He was a Post-Doctoral Fellow, followed by a Research Assistant Professor with the Department of Systems Design Engineering, University of Waterloo. He is currently a tenure-track Assistant Professor with the Department of Geomatics Engineering, University of Calgary, Calgary, AB, Canada. His work has led to significant advancements in pan-Arctic environment (i.e., sea/lake ice, arctic species, and marine oil spills) mapping, land environment (soil, crop, wildfire, forest, land cover, and land use) mapping, and hyperspectral environmental mapping and assessments. He has published various articles in high-impact remote sensing journals and conferences, such as the *Remote Sensing of Environment*, IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING, *International Journal of Applied Earth Observation and Geoinformation*, and IEEE TRANSACTIONS ON INTELLIGENT TRANSPORTATION SYSTEMS. His research is at the cutting edge of integrating machine learning, AI, and remote sensing technologies, including SAR, hyperspectral, multispectral, and LiDAR, to tackle pressing environmental challenges.

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